

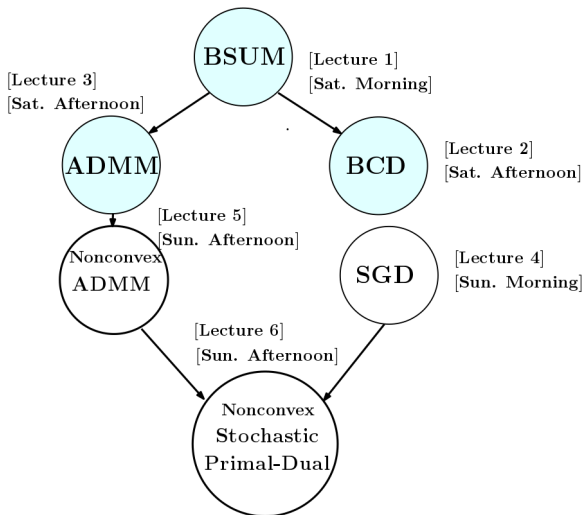
Lecture 3: Convex Primal-Dual Decomposition Methods

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Schedule (temporary)



This Time

- Introduce a family of algorithm called **A**lternating **D**irection **M**ethod of **M**ultipliers (ADMM)
- Popular algorithm for large-scale machine learning, distributed computation, etc.
- Lots of applications
- Our discussion will be based on the following review paper:
“Distributed Optimization and Statistical Learning via the Alternating Direction Method of Multipliers”, S. Boyd, N. Parikh, E. Chu, B. Peleato, J Eckstein, *Foundation and Trends in Machine Learning*, 2011

Outline

- 1 Review
- 2 Applications Overview
- 3 The Algorithm
- 4 Applications in Details
- 5 Application in B-F separation for Video Processing

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Introduction

Total citations [Cited by 1907](#)



Scholar articles [Distributed optimization and statistical learning via the alternating direction method of multipliers](#)
 S Boyd, N Parikh, E Chu, B Peleato, J Eckstein - Foundations and Trends® in Machine Learning, 2011
[Cited by 1907](#) - [Related articles](#) - [All 31 versions](#)

Total citations [Cited by 1027](#)



Scholar articles [On the Douglas—Rachford splitting method and the proximal point algorithm for maximal monotone operators](#)
 J Eckstein, DP Bertsekas - Mathematical Programming, 1992
[Cited by 1027](#) - [Related articles](#) - [All 14 versions](#)

Motivation

- **An Ancient Strategy**
- “Divide et impera” – Julius Caesar (100- 44 BC)
- “Divide and Rule” – Sun Tzu, ART OF WAR (535-470 BC)
- In mathematical terms: Split and Alternate

The Problem to be Solved

- The ADMM algorithm solves the following convex program

$$\begin{aligned} \min \quad & f(\mathbf{x}) + g(\mathbf{z}) \\ \text{s.t.} \quad & \mathbf{Ax} + \mathbf{Bz} = \mathbf{c} \\ & \mathbf{x} \in X, \quad \mathbf{z} \in Z \end{aligned}$$

- $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{z} \in \mathbb{R}^m$ are the variables
- f, g are two convex function, possible **nonsmooth**
- $\mathbf{A}, \mathbf{B}$ are two known matrices, $\mathbf{c}$ is a known vector
- Note:** Two blocks of variables; separable in the objective, coupled by a linear equation
- Main Benefit:** Capable of dealing with $\mathbf{x}$ and $\mathbf{z}$ **separately** (in a BCD manner)

Application 1: Constrained ℓ_1 problem

$$\begin{array}{ll} \min & \|\mathbf{x}\|_1 \\ \text{s.t.} & \|\mathbf{Ax} - \mathbf{b}\|^2 \leq \delta \end{array}$$

- Similar as the LASSO, but with explicit constraints on the noise magnitude
- Formulate into the ADMM form?
- Introduce a new variable $\mathbf{z}$ (**split!**), arrive at an **equivalent formulation**

$$\begin{array}{ll} \min & \|\mathbf{x}\|_1 \\ \text{s.t.} & \|\mathbf{z}\|^2 \leq \delta \\ & \mathbf{z} = \mathbf{Ax} - \mathbf{b} \end{array}$$

- Mapping back to the original problem, $f(\mathbf{x}) = \|\mathbf{x}\|_1$, $g(\mathbf{z}) = 0$
- $\mathbf{B} = -\mathbf{I}$, $Z := \{\mathbf{z} \mid \|\mathbf{z}\|^2 \leq \delta\}$, $X = \mathbb{R}^n$

Application 2: Multiple Regularizations

- Consider the following **sparse group LASSO** problem
- The problem is formulated as

$$\min \frac{1}{2} \|\mathbf{Ax} - \mathbf{b}\|^2 + \underbrace{\sum_{g=1}^G \lambda_g \|\mathbf{x}_g\|_2}_{\ell_1/\ell_2 \text{ norm}} + \underbrace{\lambda \|\mathbf{x}\|_1}_{\ell_1 \text{ norm}}$$

- Select a few groups, and select a few variables within groups
- How to deal with **two** different nonsmooth penalizations simultaneously?

Application 2: Multiple Regularizations

- Same trick, **split and alternate**
- Introduce two new variables $\mathbf{z}_1, \mathbf{z}_2$
- Perform the following reformulation

$$\min \quad \underbrace{\sum_{g=1}^G \lambda_g \|\mathbf{x}_g\|_2}_{f(\mathbf{x})} + \underbrace{\frac{1}{2} \|\mathbf{A}\mathbf{z}_2 - \mathbf{b}\|^2 + \lambda \|\mathbf{z}_1\|_1}_{g(\mathbf{z})}$$

subject to $\mathbf{z}_1 = \mathbf{x}, \quad \mathbf{z}_2 = \mathbf{x}$

- Reformulated in an ADMM form; deal with two norms separately?

Application 3: Intersection of Two Sets

- Suppose we would like to solve the following problem

$$\begin{aligned} & \text{FIND } \mathbf{x} \\ & \text{subject to } \|\mathbf{x}\|_1 \leq \delta, \quad \|\mathbf{Ax} - \mathbf{b}\|^2 \leq \epsilon \end{aligned} \quad (1)$$

- Finding the intersection of two convex sets
- We can introduce two variables $\mathbf{z}_1, \mathbf{z}_2$

$$\begin{aligned} & \text{FIND } \mathbf{x} \\ & \text{subject to } \|\mathbf{z}_1\|_1 \leq \delta, \quad \|\mathbf{z}_2\|^2 \leq \epsilon \\ & \quad \mathbf{z}_1 = \mathbf{x}, \quad \mathbf{z}_2 = \mathbf{Ax} - \mathbf{b} \end{aligned}$$

- First block $\mathbf{x}$, second block $(\mathbf{z}_1, \mathbf{z}_2)$; Fix $\mathbf{x}$, $\{\mathbf{z}_i\}$'s are **independent!**
- Dealing with each constraint **separately**

Application 4: Consensus Problem

- Consider the simple least square problem

$$\min \quad \|\mathbf{Ax} - \mathbf{b}\|^2 \quad (2)$$

- Suppose $\mathbf{A}$ is divided into N groups of rows $\mathbf{A}_1, \dots, \mathbf{A}_N$
- There are N computing nodes, each obtaining a single piece of data $\mathbf{A}_n$
- The problem can be re-written as

$$\min \quad \sum_{i=1}^N \|\mathbf{A}_i \mathbf{x} - \mathbf{b}_i\|^2 \quad (3)$$



Application 4: Consensus Problem

- Distributed computation?
- Let us introduce N variables $\mathbf{z}_1, \dots, \mathbf{z}_n$
- Consider the following reformulation

$$\begin{aligned} \min \quad & \sum_{i=1}^N \|\mathbf{A}_i \mathbf{z}_i - \mathbf{b}_i\|^2 \\ \text{subject to} \quad & \mathbf{z}_i = \mathbf{x}, \forall i \end{aligned}$$

- Two-blocks: $\mathbf{x}, (\mathbf{z}_1, \dots, \mathbf{z}_n)$
- Each $\mathbf{z}_i$ can be handled by the i -th local agent (together with $\mathbf{A}_i$)!
- Fix $\mathbf{x}$, $\{\mathbf{z}_i\}$'s are **independent**!

Application 4: Consensus Problem

- More generally, consider the following problem

$$\min \sum_i f_i(\mathbf{x}) + R(\mathbf{x}), \text{ subject to } \mathbf{x} \in X \quad (4)$$

- We can do the same procedure as before

$$\begin{aligned} \min \quad & \sum_i f_i(\mathbf{z}_i) + R(\mathbf{x}) \\ \text{subject to } & \mathbf{x} \in X, \mathbf{z}_i = \mathbf{x}, \forall i \end{aligned}$$

- Each agent handles a single function f_i
- Required to reach a **global consensus**
- Not necessarily to have the **“star network”**

Consensus SVM (from Boyd)

- Data $(\mathbf{a}_i, b_i), i = 1, \dots, N, \mathbf{a}_i \in \mathbb{R}^K, y_i \in \{-1, +1\}$
- Linear SVM

$$\min_{\mathbf{w}} \frac{1}{N} \sum_{i=1}^N \max \{0, 1 - b_i \mathbf{x}^T \mathbf{a}_i\} + \frac{\lambda}{2} \|\mathbf{x}\|_2^2 \quad (5)$$

- Example: $K = 2, N = 400$, splits into 20 groups

Note on the SVM

- The SVM problem is given by (non-separable case, absorbed x_0 into $\mathbf{x}$)

$$\begin{aligned} \min \quad & \frac{1}{2} \mathbf{x}^T \mathbf{x} + c \sum_i \xi_i \\ \text{subject to} \quad & b_i(\mathbf{a}_i^T \mathbf{x}) \geq 1 - \xi_i, \quad \xi_i \geq 0 \quad \forall i. \end{aligned} \quad (6)$$

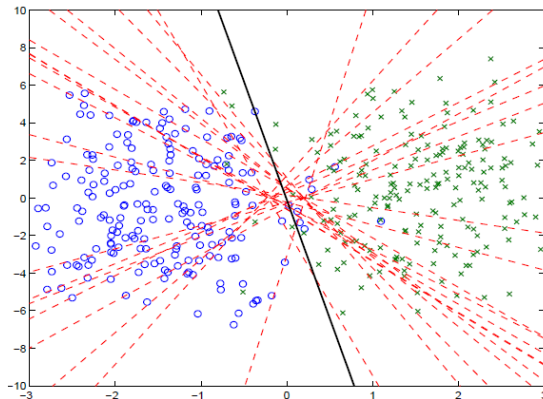
- Let's take an alternative view ...

$$\min_{\mathbf{x}} \quad \frac{1}{N} \sum_{i=1}^N \underbrace{\max\{0, 1 - b_i(\mathbf{a}_i^T \mathbf{x})\}}_{\text{hinge loss}} + \gamma \mathbf{x}^T \mathbf{x} \quad (7)$$

- Equivalent formulation

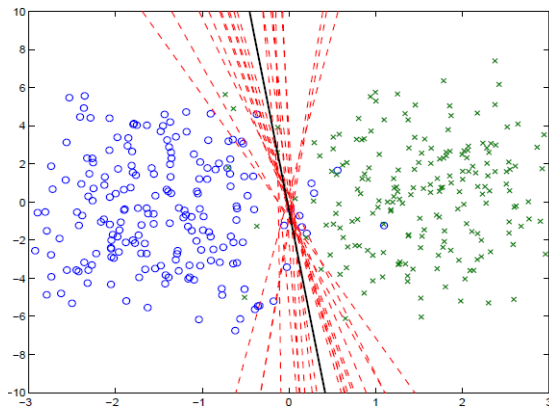
Consensus SVM (from Boyd)

Iteration 1



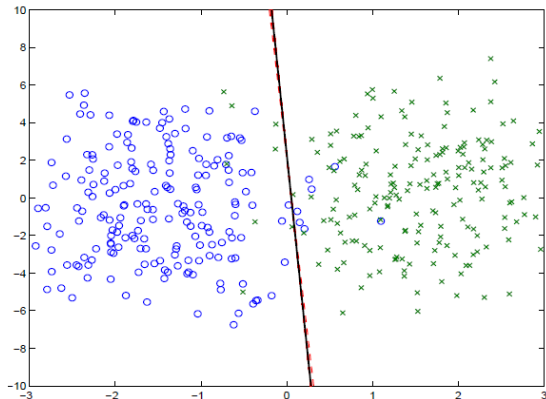
Consensus SVM (from Boyd)

Iteration 5



Consensus SVM (from Boyd)

Iteration 40



Application 5: Sharing Problem

- Consider the following **sharing problem**

$$\min \sum_{i=1}^N f_i(\mathbf{x}_i) + g \left(\sum_{i=1}^N \mathbf{x}_i \right) \quad (8)$$

- f_i local cost function; g **shared objective**
- Important class of problem, applications for example in economics
- Agents have selfish interests, but their behaviors are coupled together by some sort of “mutual interactions”

Application 5: Sharing Problem

- **First Approach** Introducing a new variable $\mathbf{z}$

$$\min \sum_{i=1}^N f_i(\mathbf{x}_i) + g(\mathbf{z})$$

$$\mathbf{z} = \sum_{i=1}^N \mathbf{x}_i$$

- **Second Approach** Introducing a set of new variable $\{\mathbf{z}_i\}$

$$\min \sum_{i=1}^N f_i(\mathbf{x}_i) + g\left(\sum_{i=1}^N \mathbf{z}_i\right)$$

$$\mathbf{z}_i = \mathbf{x}_i$$

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Review: Dual Problem for Constrained Optimization

- Primal problem (with equality constraints)

$$(P) : \min f(\mathbf{x}), \quad \text{subject to } \mathbf{Ax} = \mathbf{b}, \mathbf{x} \in X \quad (9)$$

$f(\cdot)$ is a convex function

- Consider the Lagrangian, $\mathbf{y}$ is the dual variable (multiplier)

$$L(\mathbf{x}, \mathbf{y}) = f(\mathbf{x}) + \langle \mathbf{y}, \mathbf{Ax} - \mathbf{b} \rangle$$

- Dual function

$$d(\mathbf{y}) = \min_{\mathbf{x} \in X} L(\mathbf{x}, \mathbf{y}) \leq \min_{\mathbf{Ax}=\mathbf{b}, \mathbf{x} \in X} f(\mathbf{x})$$

- Dual problem (unconstrained)

$$(D) : \max_{\mathbf{y}} d(\mathbf{y}) = \max_{\mathbf{y}} \min_{\mathbf{x} \in X} L(\mathbf{x}, \mathbf{y}) \leq \min_{\mathbf{Ax}=\mathbf{b}, \mathbf{x} \in X} f(\mathbf{x})$$

- f convex, X convex set, strong duality, $P = D$ (optimal objective equals)

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- f convex, X convex set, **strong duality**, $P = D$ (optimal objective equals)

Dual Ascent

- Dual objective is concave

$$(D) : \max_{\mathbf{y}} d(\mathbf{y}) = \max_{\mathbf{y}} \min_{\mathbf{x}} L(\mathbf{x}, \mathbf{y}) = \max_{\mathbf{y}} \min_{\mathbf{x}} f(\mathbf{x}) + \langle \mathbf{y}, \mathbf{Ax} - \mathbf{b} \rangle$$

- Gradient Ascent for the dual problem

$$\mathbf{y}^{r+1} = \mathbf{y}^r + \alpha \nabla d(\mathbf{y}^r)$$

- Side Note: apply dual ascent to SVM problem is quite standard and fast
- Will talk a bit more about dual ascent in the subsequent lectures

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Dual Ascent (cont.)

- Let $\tilde{\mathbf{x}} = \arg \min_{\mathbf{x} \in X} L(\mathbf{x}, \mathbf{y}^r)$, we know (can be verified)

$$\partial d(\mathbf{y}^r) = \mathbf{A}\tilde{\mathbf{x}} - \mathbf{b}$$

- That is, one of the subgradient for the dual function is the current constraint violation (why?)
- Dual Ascent Algorithm** (intuition?)

$$\begin{aligned}\mathbf{x}^{r+1} &= \arg \min_{\mathbf{x} \in X} L(\mathbf{x}, \mathbf{y}^r) \\ \mathbf{y}^{r+1} &= \mathbf{y}^r + \alpha (\mathbf{A}\mathbf{x}^{r+1} - \mathbf{b})\end{aligned}$$

Dual Decomposition

- If f is **separable among variables**

$$(P) : \min_{\sum_{i=1}^N \mathbf{A}_i \mathbf{x}_i = \mathbf{b}} f(\mathbf{x}) = \sum_{i=1}^N f_i(\mathbf{x}_i) \quad (10)$$

- Consider the Lagrangian

$$L(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^N L_i(\mathbf{x}_i, \mathbf{y}) = \sum_{i=1}^N f_i(\mathbf{x}_i) + \langle \mathbf{y}, \mathbf{A}_i \mathbf{x}_i - \mathbf{b} \rangle \quad (11)$$

- Dual ascent algorithm (a.k.a. dual decomposition)

$$\begin{aligned} \mathbf{x}_i^{r+1} &= \arg \min_{\mathbf{x}_i \in X_i} L_i(\mathbf{x}_i, \mathbf{y}^r), \quad \forall i \\ \mathbf{y}^{r+1} &= \mathbf{y}^r + \alpha \left(\sum_{i=1}^N \mathbf{A}_i \mathbf{x}_i^{r+1} - \mathbf{b} \right) \end{aligned}$$

- Large-scale problem: Split variables, update $\mathbf{x}_i$ in **parallel**

Convergence of Dual Decomposition

- Assumption for convergence
 - 1 Dual function $d(\mathbf{y})$ has **Lipschitz continuous gradient** (with constant L)
 - 2 The stepsize $\alpha \leq \frac{1}{L}$
- (1) requires that problem (P) is **strongly convex** with constant at least $\frac{1}{L}$

Method of Multipliers

- Primal problem (with equality constraints)

$$(P) : \min f(\mathbf{x}), \quad \text{subject to } \mathbf{Ax} = \mathbf{b}, \mathbf{x} \in X \quad (12)$$

$f(\cdot)$ is a convex function

- Consider the **Augmented Lagrangian**, $\mathbf{y}$ is the dual variable (multiplier)

$$L(\mathbf{x}, \mathbf{y}) = f(\mathbf{x}) + \langle \mathbf{y}, \mathbf{Ax} - \mathbf{b} \rangle + \frac{\rho}{2} \|\mathbf{Ax} - \mathbf{b}\|^2$$

- $\rho > 0$ is penalty parameter
- Quadratic penalty term penalizes constraint violation

Method of Multipliers

- **Method of Multipliers** (intuition?)

$$\mathbf{x}^{r+1} = \arg \min_{\mathbf{x} \in X} L_{\rho}(\mathbf{x}, \mathbf{y}^r)$$

$$\mathbf{y}^{r+1} = \mathbf{y}^r + \rho (\mathbf{A}\mathbf{x}^{r+1} - \mathbf{b})$$

Method of Multipliers

- **Good**

- 1 Weak Assumptions: f only needs to be convex
- 2 Dual function has Lipschitz continuous gradient

- **Bad**

- 1 Objective **not separable** anymore
- 2 Dual decomposition no longer works

- ADMM combines

- 1 Method of multipliers: Use **augmented Lagrangian**
- 2 Dual decomposition: Can do variable splitting

The ADMM Algorithm

- Consider the Equality-Constrained convex problem

$$\min_{\mathbf{x} \in X, \mathbf{z} \in Z} f(\mathbf{x}) + g(\mathbf{z}), \quad \text{s.t. } \mathbf{Ax} + \mathbf{Bz} = \mathbf{c}$$

- Augmented Lagrangian

$$L_\rho(\mathbf{x}, \mathbf{z}; \mathbf{y}) = f(\mathbf{x}) + g(\mathbf{z}) + \langle \mathbf{y}, \mathbf{Ax} + \mathbf{Bz} - \mathbf{c} \rangle + \frac{\rho}{2} \|\mathbf{Ax} + \mathbf{Bz} - \mathbf{c}\|^2$$

- Method of multipliers

$$\begin{aligned} (\mathbf{x}^{r+1}, \mathbf{z}^{r+1}) &= \arg \min_{\mathbf{x}, \mathbf{z}} L_\rho(\mathbf{x}, \mathbf{z}; \mathbf{y}^r) \\ \mathbf{y}^{r+1} &= \mathbf{y}^r + \rho (\mathbf{Ax}^{r+1} + \mathbf{Bz}^{r+1} - \mathbf{c}) \end{aligned}$$

The ADMM Algorithm

The steps of the ADMM Algorithm is given below

$$\mathbf{x}^{r+1} = \arg \min_{\mathbf{x} \in X} L_{\rho}(\mathbf{x}, \mathbf{z}^r; \mathbf{y}^r)$$

$$\mathbf{z}^{r+1} = \arg \min_{\mathbf{z} \in Z} L_{\rho}(\mathbf{x}^{r+1}, \mathbf{z}; \mathbf{y}^r)$$

$$\mathbf{y}^{r+1} = \mathbf{y}^r + \rho (\mathbf{A}\mathbf{x}^{r+1} + \mathbf{B}\mathbf{z}^{r+1} - \mathbf{c})$$

- Divide and conquer: Optimize $\mathbf{x}$ and $\mathbf{z}$ once (coordinate descent on L_{ρ}), then update the dual variable
- The primal problem is no longer solved exactly (where the efficiency comes from)

The ADMM Algorithm

- One can view it as updating **three variables** (2 primal, 1 dual) alternately
- We can switch the role of two primal variables $\mathbf{x}$ and $\mathbf{z}$
- There is no restriction on the stepsize ρ , at long as $\rho > 0$

Optimality Conditions (READ)

- Lagrangian

$$L(\mathbf{x}, \mathbf{z}; \mathbf{y}) = f(\mathbf{x}) + g(\mathbf{z}) + \langle \mathbf{y}, \mathbf{Ax} + \mathbf{Bz} - \mathbf{c} \rangle$$

- KKT condition (suppose X, Z are both the **whole space**, no other constraints on $\mathbf{x}, \mathbf{z}$)

$$-\mathbf{A}^T \mathbf{y}^* \in \partial f(\mathbf{x}^*), \quad -\mathbf{B}^T \mathbf{y}^* \in \partial g(\mathbf{z}^*)$$

$$\mathbf{Ax}^* + \mathbf{Bz}^* - \mathbf{c} = 0$$

- ADMM updates (optimality condition at each iteration)

$$-\mathbf{A}^T \mathbf{y}^{r+1} + \mathbf{A}^T (\mathbf{Bz}^{t+1} - \mathbf{Bz}^t) \in \partial f(\mathbf{x}^{r+1})$$

$$-\mathbf{B}^T \mathbf{y}^{r+1} \in \partial g(\mathbf{z}^{r+1})$$

- Optimality is achieved if the following are satisfied:

$$\mathbf{A}^T (\mathbf{Bz}^{t+1} - \mathbf{Bz}^t) = 0$$

$$\mathbf{Ax}^{t+1} + \mathbf{Bz}^{t+1} - \mathbf{c} = 0$$

Convergence

- The ADMM converges under very mild condition
- Let us define the **residue** as

$$\mathbf{r}^r = \mathbf{A}\mathbf{x}^r + \mathbf{B}\mathbf{z}^r - \mathbf{c}$$

- **Claim:** Suppose that the problem is convex and **feasible**, then the following is true
 - 1 **Residue convergence:** $\mathbf{r}^k \rightarrow 0$ as $k \rightarrow \infty$
 - 2 **Objective convergence:** $f(\mathbf{x}^k) + g(\mathbf{z}^k) \rightarrow p^*$ as $k \rightarrow \infty$
 - 3 **Dual variable convergence:** $\mathbf{y}^k \rightarrow \mathbf{y}^*$ as $k \rightarrow \infty$, where $\mathbf{y}^*$ is the optimal dual solution

Convergence Rate

- Let $\bar{\mathbf{x}}^T = \frac{1}{T} \sum_{r=1}^T \mathbf{x}^t$ (averaging over all past iterates)
- Let $\bar{\mathbf{z}}^T = \frac{1}{T} \sum_{r=1}^T \mathbf{z}^t$ (averaging over all past iterates)
- Convergence rate for the objective

$$f(\bar{\mathbf{x}}^T) + g(\bar{\mathbf{z}}^T) - [f(\bar{\mathbf{x}}^*) + g(\bar{\mathbf{z}}^*)] \leq \mathcal{O}\left(\frac{1}{T}\right)$$

- Convergence rate for optimality conditions

$$\|\mathbf{Ax}^T + \mathbf{Bz}^T - \mathbf{c}\|^2 + \|\mathbf{Bz}^T - \mathbf{Bz}^{T-1}\|^2 \leq \mathcal{O}\left(\frac{1}{T}\right)$$

Next

- Go over a few applications, and see how ADMM iteration works
- Go over in detail a video background/foreground separation problem
- See how the algorithm works in practice

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- Similar as the LASSO, but with explicit constraints on the noise magnitude
- Formulate into the ADMM form?
- Introduce new variables $\mathbf{z}_1, \mathbf{z}_2$

$$\begin{array}{ll} \min & \|\mathbf{x}\|_1 \\ \text{s.t.} & \|\mathbf{z}_1\|^2 \leq \delta \\ & \mathbf{z}_1 = \mathbf{Ax}_2 - \mathbf{b}, \mathbf{z}_2 = \mathbf{x} \end{array}$$

ADMM for Constrained ℓ_1 problem

- Formulate the augmented Lagrangian

$$L(\mathbf{x}, \mathbf{z}; \mathbf{y}) = \|\mathbf{x}\|_1 + \langle \mathbf{y}_1, \mathbf{z}_1 - \mathbf{A}\mathbf{z}_2 + \mathbf{b} \rangle + \frac{\rho}{2} \|\mathbf{z}_1 - \mathbf{A}\mathbf{z}_2 + \mathbf{b}\|^2 \\ + \langle \mathbf{y}_2, \mathbf{z}_2 - \mathbf{x} \rangle + \frac{\rho}{2} \|\mathbf{z}_2 - \mathbf{x}\|^2$$

- Put $(\mathbf{x}, \mathbf{z}_1)$ into one group (they are not “coupled” in the objective)
- The $(\mathbf{x}, \mathbf{z}_1)$ -step (proximity step + projection onto a ball)

$$\mathbf{x}^{r+1} = \arg \min_{\mathbf{x}} \|\mathbf{x}\|_1 + \langle \mathbf{y}_2^r, \mathbf{z}_2^r - \mathbf{x} \rangle + \frac{\rho}{2} \|\mathbf{z}_2^r - \mathbf{x}\|^2 \\ \mathbf{z}_1^{r+1} = \arg \min_{\|\mathbf{z}_1\|^2 \leq \delta} \langle \mathbf{y}_1^r, \mathbf{z}_1 - \mathbf{A}\mathbf{z}_2^r + \mathbf{b} \rangle + \frac{\rho}{2} \|\mathbf{z}_1 - \mathbf{A}\mathbf{z}_2^r + \mathbf{b}\|^2$$

- The $\mathbf{z}_2$ -step (can be solved in closed form)

$$\mathbf{z}_2^{t+1} = \arg \min_{\mathbf{z}_2} \langle \mathbf{y}_1^r, -\mathbf{A}\mathbf{z}_2 \rangle + \frac{\rho}{2} \|\mathbf{z}_1^{r+1} - \mathbf{A}\mathbf{z}_2 - \mathbf{b}\|^2 + \langle \mathbf{y}_2^r, \mathbf{z}_2 \rangle + \frac{\rho}{2} \|\mathbf{z}_2 - \mathbf{x}^{r+1}\|^2$$

- Then update $\mathbf{y}_1, \mathbf{y}_2$ (easy)

Note: How to solve x problem? The Soft-Thresholding

- Consider the following scalar optimization problem

$$\min \frac{1}{2}|x - b|^2 + \lambda|x| \quad (P)$$

where $\lambda > 0$ is some coefficient, $x \in \mathbb{R}$

- First order optimality condition $0 \in \partial f(x)$ (recall subgradient of $|x|$)

$$\partial f(x) = \begin{cases} (x - b) - \lambda & \text{if } x < 0 \\ (-b) + \lambda\epsilon & \text{if } x = 0 \\ (x - b) + \lambda & \text{if } x > 0 \end{cases} \quad (13)$$

where $\epsilon \in [-1, 1]$

- The solution becomes (interpretation? What happens when λ is large?)

$$x^* = \begin{cases} b + \lambda & \text{if } b + \lambda < 0 \\ 0 & \text{if } |b| \leq \lambda \\ b - \lambda & \text{if } b - \lambda > 0 \end{cases} \quad (14)$$

ADMM for Constrained ℓ_1 problem - Summary

$$\begin{array}{ll} \min & \|\mathbf{x}\|_1 \\ \text{s.t.} & \|\mathbf{Ax} - \mathbf{b}\|^2 \leq \delta \end{array}$$

- **Main difficulty**: objective nonsmooth + difficult constraint
- First step, introduce new variables to split these two things
- Apply ADMM, each “difficulty” is dealt with **separably**
- **Divide and Conquer!**

Application 2: Multiple Regularizations

- Consider the following **sparse group LASSO** problem
- The problem is formulated as

$$\min \frac{1}{2} \|\mathbf{Ax} - \mathbf{b}\|^2 + \underbrace{\sum_{g=1}^G \lambda_g \|\mathbf{x}_g\|_2}_{\ell_1/\ell_2 \text{ norm}} + \underbrace{\lambda \|\mathbf{x}\|_1}_{\ell_1 \text{ norm}}$$

- Select a few groups, and select a few variables within groups
- How to deal with two different norms simultaneously?

Application 2: Multiple Regularizations

- Same trick, do the variable splitting
- Introduce new variable $\mathbf{z}_1, \mathbf{z}_2$
- Perform the following reformulation

$$\begin{aligned}
 \min \quad & \frac{1}{2} \|\mathbf{A}\mathbf{z}_2 - \mathbf{b}\|^2 + \sum_{g=1}^G \lambda_g \|\mathbf{x}_g\|_2 + \lambda \|\mathbf{z}_1\|_1 \\
 \text{subject to} \quad & \mathbf{z}_1 = \mathbf{x}, \quad \mathbf{z}_2 = \mathbf{x}
 \end{aligned} \tag{15}$$

- Reformulated in an ADMM form; deal with two norms separately?

ADMM for Multiple Regularizations

- Again let's first formulate the augmented Lagrangian

$$L(\mathbf{x}, \mathbf{z}; \mathbf{y}) = \frac{1}{2} \|\mathbf{A}\mathbf{z}_1 - \mathbf{b}\|^2 + \sum_{g=1}^G \lambda_g \|\mathbf{x}_g\|_2 + \lambda \|\mathbf{z}_2\|_1 \\ + \langle \mathbf{y}_1, \mathbf{z}_1 - \mathbf{x} \rangle + \frac{\rho}{2} \|\mathbf{z}_1 - \mathbf{x}\|^2 + \langle \mathbf{y}_2, \mathbf{z}_2 - \mathbf{x} \rangle + \frac{\rho}{2} \|\mathbf{z}_2 - \mathbf{x}\|^2$$

- $\mathbf{x}$ is one variable block, $(\mathbf{z}_1, \mathbf{z}_2)$ is a variable block
- $\mathbf{x}$ subproblem (proximal operator for group-Lasso)

$$\mathbf{x}^{r+1} = \arg \min \sum_{g=1}^G \lambda_g \|\mathbf{x}_g\|_2 + \frac{\rho}{2} \left\| \mathbf{x} - \mathbf{z}_1^r - \frac{1}{\rho} \mathbf{y}_1^r \right\|^2 + \frac{\rho}{2} \left\| \mathbf{x} - \mathbf{z}_2^r - \frac{1}{\rho} \mathbf{y}_2^r \right\|^2$$

- $\mathbf{z}_1$ subproblem easy (quadratic, unconstrained)
- $\mathbf{z}_2$ subproblem is proximal operator for Lasso

Application 3: Intersection of Two Sets

- Suppose we would like to find a vector that

$$\begin{aligned} & \text{FIND} \\ & \text{subject to } \| \mathbf{x} \|_1 \leq \delta, \quad \| \mathbf{Ax} - \mathbf{b} \|^2 \leq \epsilon \end{aligned} \quad (16)$$

- Finding a intersection of two convex sets
- We can introduce two variables $\mathbf{z}_1, \mathbf{z}_2$

$$\begin{aligned} & \text{FIND} \\ & \text{subject to } \| \mathbf{z}_1 \|_1 \leq \delta, \quad \| \mathbf{z}_2 \|^2 \leq \epsilon \\ & \quad \mathbf{z}_1 = \mathbf{x}, \quad \mathbf{z}_2 = \mathbf{Ax} - \mathbf{b} \end{aligned}$$

- First block $\mathbf{x}$, second block $(\mathbf{z}_1, \mathbf{z}_2)$
- Dealing with each constraint **separately**

Application 3: Intersection of Two Sets

- The augmented Lagrangian

$$L(\mathbf{x}, \mathbf{z}; \mathbf{y}) = \langle \mathbf{z}_1 - \mathbf{x}, \mathbf{y}_1 \rangle + \frac{\rho}{2} \|\mathbf{z}_1 - \mathbf{x}\|^2 \\ + \langle \mathbf{z}_2 - \mathbf{Ax} + \mathbf{b}, \mathbf{y}_2 \rangle + \frac{\rho}{2} \|\mathbf{z}_2 - \mathbf{Ax} + \mathbf{b}\|^2$$

- $\mathbf{z}_1$ subproblem (projection on ℓ_1 ball, easy, see [Here](#))

$$\min \quad \langle \mathbf{z}_1, \mathbf{y}_1^r \rangle + \frac{\rho}{2} \|\mathbf{z}_1 - \mathbf{x}^r\|^2, \quad \text{subject to } \|\mathbf{z}_1\|_1 \leq \delta$$

- $\mathbf{z}_2$ subproblem (projection on ℓ_2 ball, easy)

$$\min \quad \langle \mathbf{z}_2, \mathbf{y}_2^r \rangle + \frac{\rho}{2} \|\mathbf{z}_2 - \mathbf{Ax}^r + \mathbf{b}\|^2, \quad \text{subject to } \|\mathbf{z}_2\|_2^2 \leq \epsilon$$

- $\mathbf{x}$ subproblem, simple unconstrained quadratic problem, easy

Application 4: Consensus Problem

- Consider the following problem

$$\min \sum_{i=1}^N f_i(\mathbf{x}) + R(\mathbf{x}), \text{ subject to } \mathbf{x} \in X \quad (17)$$

- We can do the same procedure as before

$$\begin{aligned} \min \quad & \sum_{i=1}^N f_i(\mathbf{z}_i) + R(\mathbf{x}) \\ \text{subject to } & \mathbf{x} \in X, \mathbf{z}_i = \mathbf{x}, \forall i \end{aligned}$$

- Each agent handles a single function f_i

ADMM for Consensus Problem

- Write down the augmented Lagrangian

$$L(\mathbf{x}, \mathbf{z}; \mathbf{y}) = \sum_{i=1}^N f_i(\mathbf{z}_i) + R(\mathbf{x}) + \sum_{i=1}^N \langle \mathbf{y}_i, \mathbf{z}_i - \mathbf{x} \rangle + \frac{\rho}{2} \|\mathbf{z}_i - \mathbf{x}\|^2$$

- $\mathbf{x}$ subproblem (proximity operator, at the control center)

$$\mathbf{x}^{r+1} = \arg \min R(\mathbf{x}) + \sum_{i=1}^N \langle \mathbf{y}_i^r, -\mathbf{x} \rangle + \frac{\rho}{2} \|\mathbf{z}_i^r - \mathbf{x}\|^2$$

- For each agent, $\mathbf{z}_i$ problem (at the agent)

$$\mathbf{z}_i^{r+1} = \arg \min f_i(\mathbf{z}_i) + \langle \mathbf{y}_i^r, \mathbf{z}_i \rangle + \frac{\rho}{2} \|\mathbf{z}_i - \mathbf{x}^{r+1}\|^2$$

- Update the dual variables (at the control center)

$$\mathbf{y}_i^{r+1} = \mathbf{y}_i^r + \rho(\mathbf{z}_i^{r+1} - \mathbf{x}^{r+1}), \forall i$$

ADMM for Consensus Problem (cont.)

- **Special Case:** Without the regularization term, we have

$$L(\mathbf{x}, \mathbf{z}; \mathbf{y}) = \sum_{i=1}^N f_i(\mathbf{z}_i) + \sum_{i=1}^N \langle \mathbf{y}_i, \mathbf{z}_i - \mathbf{x} \rangle + \frac{\rho}{2} \|\mathbf{z}_i - \mathbf{x}\|^2$$

- For each agent, $\mathbf{z}_i$ problem (at the agent)

$$\mathbf{z}_i^{r+1} = \arg \min f_i(\mathbf{z}_i) + \langle \mathbf{y}_i^r, \mathbf{z}_i \rangle + \frac{\rho}{2} \|\mathbf{z}_i - \mathbf{x}^r\|^2$$

- $\mathbf{x}$ subproblem (at the control center)

$$\mathbf{x}^{r+1} = \arg \min \sum_{i=1}^N \langle \mathbf{y}_i^r, -\mathbf{x} \rangle + \frac{\rho}{2} \|\mathbf{z}_i^{r+1} - \mathbf{x}\|^2 = \frac{1}{N} \sum_{i=1}^N (\mathbf{z}_i^{r+1} + \mathbf{y}_i^r / \rho)$$

- Update the dual variables (at the control center)

$$\mathbf{y}_i^{r+1} = \mathbf{y}_i^r + \rho(\mathbf{z}_i^{r+1} - \mathbf{x}^{r+1}), \forall i$$

- One can show (make sense ?)

$$\sum_{i=1}^N \mathbf{y}_i^{r+1} = 0 \longrightarrow \mathbf{x}^{r+1} = \frac{1}{N} \sum_{i=1}^N \mathbf{z}_i^{r+1}$$

ADMM for Consensus Problem (cont.)

- Let $\bar{\mathbf{z}}^{r+1} = \frac{1}{N} \sum_{i=1}^N \mathbf{z}_i^{r+1}$
- ADMM updates reduces to:

$$\mathbf{z}_i^{r+1} = \arg \min f_i(\mathbf{z}_i) + \langle \mathbf{y}_i^r, \mathbf{z}_i \rangle + \frac{\rho}{2} \|\mathbf{z}_i - \bar{\mathbf{z}}^r\|^2 \quad \forall i$$

$$\mathbf{y}_i^{r+1} = \mathbf{y}_i^r + \rho(\mathbf{z}_i^{r+1} - \bar{\mathbf{z}}^{r+1}), \quad \forall i$$

- Parallel execution by “gather-scatter”
 - Gather** $\mathbf{z}_i^r$ and average to get $\bar{\mathbf{z}}^r$
 - Scatter** the average $\bar{\mathbf{z}}^r$ to processors
 - Compute** $\mathbf{z}_i^{r+1}$ locally (in parallel)
 - Compute** $\mathbf{y}_i^{r+1}$ locally (in parallel)

Example: Consensus LASSO

- Lasso with consensus optimization (divide data by training samples)

$$\min_{\mathbf{x}} \frac{1}{2} \sum_{i=1}^N \|\mathbf{a}_i^T \mathbf{x} - b_i\|^2 + \lambda \|\mathbf{x}\|_1 = \min_{\mathbf{z}_i = \mathbf{x}} \frac{1}{2} \sum_{i=1}^N \|\mathbf{a}_i^T \mathbf{z}_i - b_i\|^2 + \lambda \|\mathbf{x}\|_1$$
- Augmented Lagrangian

$$L = \sum_{i=1}^N \frac{1}{2} \|\mathbf{a}_i^T \mathbf{z}_i - b_i\|^2 + \lambda \|\mathbf{x}\|_1 + \langle \mathbf{y}_i, \mathbf{z}_i - \mathbf{x} \rangle + \frac{\rho}{2} \|\mathbf{z}_i - \mathbf{x}\|^2$$

- ADMM updates

$$\mathbf{z}_i^{r+1} = \frac{1}{2} \|\mathbf{a}_i^T \mathbf{z}_i - b_i\|^2 + \langle \mathbf{y}_i^r, \mathbf{z}_i \rangle + \frac{\rho}{2} \|\mathbf{z}_i - \mathbf{x}^r\|^2, \quad \forall i$$

$$\mathbf{x}^{r+1} = \lambda \|\mathbf{x}\|_1 + \sum_{i=1}^N \langle \mathbf{y}_i^r, -\mathbf{x} \rangle + \frac{\rho}{2} \|\mathbf{z}_i^{r+1} - \mathbf{x}\|^2,$$

$$\mathbf{y}_i^{r+1} = \mathbf{y}_i^r + \rho(\mathbf{z}_i^{r+1} - \mathbf{x}^{r+1}), \quad \forall i$$

Example: Consensus LASSO (from Boyd)

- Example with $\mathbf{A} \in \mathbb{R}^{400,000 \times 8000}$, roughly *30GB* of data
- Split into *80* groups across *10* (each 8-core) machines
- Written in C using MPI, on Amazon EC2
- ADMM running time: *0.5-2* seconds
- Total running time, including loading data, etc, *5-6* mins.

Consensus SVM (from Boyd)

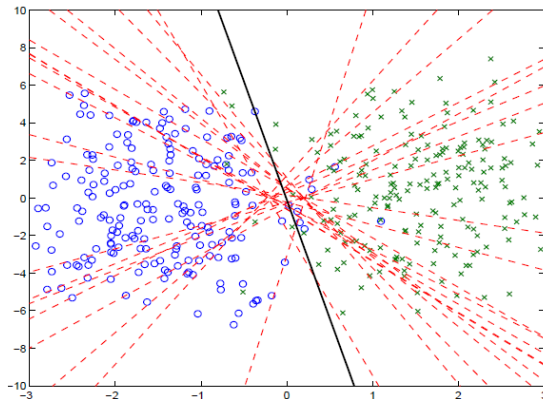
- Data $(\mathbf{a}_i, b_i), i = 1, \dots, N, \mathbf{a}_i \in \mathbb{R}^K, y_i \in \{-1, +1\}$
- Linear SVM

$$\min_{\mathbf{w}} \frac{1}{N} \sum_{i=1}^N \max \{0, 1 - b_i \mathbf{x}^T \mathbf{a}_i\} + \frac{\lambda}{2} \|\mathbf{x}\|_2^2 \quad (18)$$

- Example: $K = 2, N = 400$, splits into 20 groups

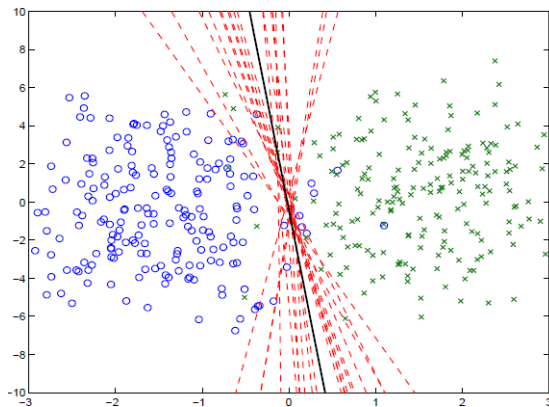
Consensus SVM (from Boyd)

Iteration 1



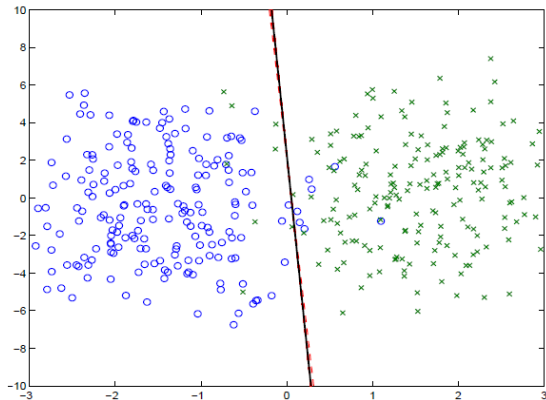
Consensus SVM (from Boyd)

Iteration 5



Consensus SVM (from Boyd)

Iteration 40



Outline

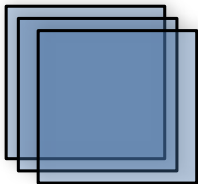
- 1 Review
- 2 Applications Overview
- 3 The Algorithm
- 4 Applications in Details
- 5 Application in B-F separation for Video Processing

Application: B-F separation and video processing

- This application is related to anomaly detection
- **Objective.** Find unusual and rare events from a large amount of data
- Lots of applications, data analysis, imaging, cybersecurity
- **Ways.** Utilizing the sparsity of the “anomaly”
- We will illustrate this using the application of background foreground separation in video processing

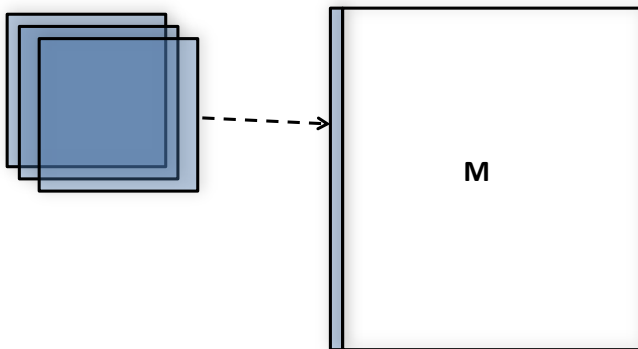
Application: B-F separation

- An application on foreground separation from noisy surveillance video
- Suppose we have a sequence of video frames $\{\mathbf{M}^i\}_{i=1}^n$
- First we do preprocessing:



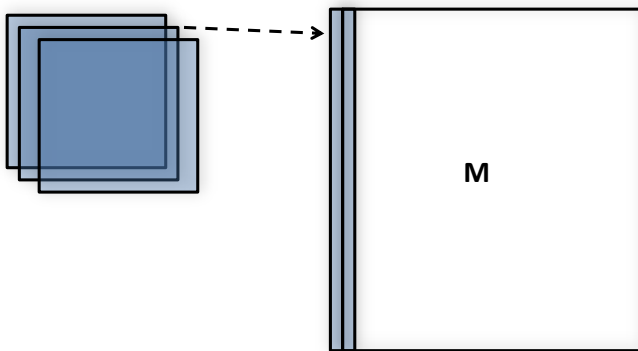
Application: B-F separation

- An application on foreground separation from noisy surveillance video
- Suppose we have a sequence of video frames $\{\mathbf{M}^i\}_{i=1}^n$
- First we do preprocessing:



Application: B-F separation

- An application on foreground separation from noisy surveillance video
- Suppose we have a sequence of video frames $\{\mathbf{M}^i\}_{i=1}^n$
- First we do preprocessing:



Application: B-F separation

- We make each column of $\mathbf{M}$ becomes **one frame** of video
- **Objective:** Decompose it into three parts:
 - 1 $\mathbf{L}$ is the background (**low rank**)
 - 2 $\mathbf{S}$ is the moving foreground (**sparse**)
 - 3 $\mathbf{Z}$ is the noise (**small magnitude**)
- Why low rank matrix represents background? Changes slowly; consider a matrix with identical columns, the rank is **1**
- **Note:** No “label”, unsupervised learning!
- **Main Idea:** Promote these properties by penalizing using different costs:
 $(\min \text{Rank}(\mathbf{L}) + \text{Density}(\mathbf{S}) + \text{Magnitude}(\mathbf{Z}))$

Application: B-F separation

$$\begin{array}{ll} \min & \|\mathbf{L}\|_* + \tau_1 \|\mathbf{S}\|_1 + \tau_2 \|\mathbf{Z}\|_F^2 \\ \text{subject to} & \mathbf{L} + \mathbf{S} + \mathbf{Z} = \mathbf{M} \end{array}$$

- 250×135 pixels per frame, 50 frames, over 1 million variables!!
- Similar idea applies to
 - 1 Analyzing power network security (intrusion detection)
 - 2 Data cleansing (getting rid of the data outliers)
 - 3 Network data flow with intrusion detection

Problem Setup

$$\begin{array}{ll} \min & \|\mathbf{L}\|_* + \tau \|\mathbf{S}\|_1 \\ \text{subject to} & \|(\mathbf{L} + \mathbf{S} - \mathbf{M})(\Omega)\|_F^2 \leq \mu \end{array}$$

- Consider a slightly different video F-B separation problem
- Minimize low rank plus the sparse part
- Ω is the set of **observed pixels** (say 80% of the original pixels)
- $\mathbf{X}(\Omega)$ means the submatrix of $\mathbf{X}$ defined by the index set Ω
- The constraint means the difference between the reconstructed frames and the original frame **on the observed pixels** is small
- τ and μ are two known numbers (parameters of the problem)

ADMM Formulation

$$\begin{aligned} \min \quad & \|\mathbf{L}\|_* + \tau \|\mathbf{S}\|_1 \\ \text{subject to} \quad & \|(\mathbf{L} + \mathbf{S} - \mathbf{M})(\Omega)\|_F^2 \leq \mu \end{aligned}$$

- Introduce a new variable $\mathbf{Z}$, consider

$$\begin{aligned} \min \quad & \|\mathbf{L}\|_* + \tau \|\mathbf{S}\|_1 \\ \text{subject to} \quad & \|\mathbf{L} - \mathbf{Z}(\Omega)\|_F^2 \leq \mu \\ & \mathbf{L} + \mathbf{S} + \mathbf{Z} = \mathbf{M} \end{aligned}$$

- The augmented Lagrangian

$$L(\mathbf{L}, \mathbf{S}, \mathbf{Z}; \Lambda) = \|\mathbf{L}\|_* + \tau \|\mathbf{S}\|_1 + \langle \Lambda, \mathbf{L} + \mathbf{S} + \mathbf{Z} - \mathbf{M} \rangle + \frac{\beta}{2} \|\mathbf{L} + \mathbf{S} + \mathbf{Z} - \mathbf{M}\|^2$$

The ADMM: Z-subproblem

- The augmented Lagrangian

$$L(\mathbf{L}, \mathbf{S}, \mathbf{Z}; \mathbf{\Lambda}) = \|\mathbf{L}\|_* + \tau \|\mathbf{S}\|_1 + \langle \mathbf{\Lambda}, \mathbf{L} + \mathbf{S} + \mathbf{Z} - \mathbf{M} \rangle + \frac{\beta}{2} \|\mathbf{L} + \mathbf{S} + \mathbf{Z} - \mathbf{M}\|^2$$

- Consider the $\mathbf{Z}$ subproblem
- Define $\mathbf{T} = \mathbf{M} + \mathbf{\Lambda}/\beta - \mathbf{L} - \mathbf{S}$
- The $\mathbf{Z}$ -subproblem becomes

$$\min_{\mathbf{Z}} \|\mathbf{Z} - \mathbf{T}\|^2, \quad \text{subject to } \|\mathbf{Z}(\Omega)\|_F^2 \leq \mu$$

- The solution (again a projection operator)

$$\begin{aligned} \mathbf{Z}(\Omega) &= \mathbf{T}(\Omega) \times \min(\sqrt{\mu}, \|\mathbf{T}(\Omega)\|_F) / \|\mathbf{T}(\Omega)\|_F \\ \mathbf{Z}(\bar{\Omega}) &= \mathbf{T}(\bar{\Omega}) \quad (\text{don't care about the rest pixels}) \end{aligned}$$

- Here $\mathbf{Z}(\Omega)$ means the components of $\mathbf{Z}$ corresponding to the set Ω

The ADMM: S-subproblem

- The augmented Lagrangian

$$L(\mathbf{L}, \mathbf{S}, \mathbf{Z}; \mathbf{\Lambda}) = \|\mathbf{L}\|_* + \tau \|\mathbf{S}\|_1 + \langle \mathbf{\Lambda}, \mathbf{L} + \mathbf{S} + \mathbf{Z} - \mathbf{M} \rangle + \frac{\beta}{2} \|\mathbf{L} + \mathbf{S} + \mathbf{Z} - \mathbf{M}\|^2$$

- Consider the S subproblem
- Define $\mathbf{T} = \mathbf{M} + \mathbf{\Lambda}/\beta - \mathbf{L} - \mathbf{Z}$
- The S-subproblem becomes

$$\min_{\mathbf{S}} \frac{\beta}{2} \|\mathbf{S} - \mathbf{T}\|^2 + \tau \|\mathbf{S}\|_1$$

- Solution:** Soft-thresholding $\mathbf{S} = \text{Soft}\left(\mathbf{T}, \frac{\tau}{\beta}\right)$

The ADMM: $\mathbf{L}$ -subproblem

- The augmented Lagrangian

$$L(\mathbf{L}, \mathbf{S}, \mathbf{Z}; \mathbf{\Lambda}) = \|\mathbf{L}\|_* + \tau \|\mathbf{S}\|_1 + \langle \mathbf{\Lambda}, \mathbf{L} + \mathbf{S} + \mathbf{Z} - \mathbf{M} \rangle + \frac{\beta}{2} \|\mathbf{L} + \mathbf{S} + \mathbf{Z} - \mathbf{M}\|^2$$

- Consider the $\mathbf{L}$ subproblem
- Define $\mathbf{T} = \mathbf{M} + \mathbf{\Lambda}/\beta - \mathbf{S} - \mathbf{Z}$
- The $\mathbf{L}$ -subproblem becomes

$$\min_{\mathbf{L}} \frac{1}{2} \|\mathbf{L} - \mathbf{T}\|^2 + \frac{1}{\beta} \|\mathbf{L}\|_*$$

- Solution:** Soft-thresholding on the singular values of $\mathbf{T}$

The ADMM: The dual Step

- The final step is to update the dual variable

$$\boldsymbol{\Lambda} = \boldsymbol{\Lambda} + \beta(\mathbf{L} + \mathbf{S} + \mathbf{Z} - \mathbf{M})$$

- When to stop? Two criteria

- The following quantity is small enough

$$\max \{ \|\mathbf{L}^{r+1} - \mathbf{L}^r\| / (1 + \|\mathbf{L}^{r+1}\|), \|\mathbf{S}^{r+1} - \mathbf{S}^r\| / (1 + \|\mathbf{S}^{r+1}\|) \}$$

- The following quantity is small enough

$$|f(\mathbf{L}^{r+1}, \mathbf{S}^{r+1}) - f(\mathbf{L}^r, \mathbf{S}^r)| / |f(\mathbf{L}^{r+1}, \mathbf{S}^{r+1})|$$